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Customer Engagement & Product Utilization Analytics for Retention Strategy

Research paper | Finance Analyst Internship project | Dataset: European Bank customer file (10,000 customers, 2025 snapshot)

1. Executive Summary

This study asks whether how customers behave with the bank (engagement and product use) explains churn better than who they are or how much money they hold. Across 10,000 customers, 2,037 left, an overall churn rate of **20.37%**.

Headline findings

- **Engagement matters.** Inactive members churn at 26.85% versus 14.27% for active members, an Engagement Retention Ratio of **1.88x**. Roughly half the base (48.5%) is inactive.
- **Product depth helps, but only up to two products.** Two-product customers churn at just 7.58%, against 27.71% for single-product customers. Customers with three products churn at 82.71% and all 60 customers with four products left.
- **High balances do not guarantee loyalty.** Customers in the top balance quartile churn more (23.68%) than the rest (19.27%). Inactive high-balance customers churn at **30.47%**, the highest of any profile, and hold about €185.6 million.
- **The stickiest customers are active and hold exactly two products.** This group (2,446 customers) churns at 5.56%.
- **Credit card ownership is not a retention lever.** Churn is 20.18% for card holders and 20.81% for non-holders.
- **Silent churn exposure is large.** 867 inactive high-balance customers have not yet left but hold about €128.8 million. They are the priority group for outreach.

Recommended actions:

(1) launch a re-activation programme for inactive high-balance customers;

(2) steer single-product customers towards a second product, not a third or fourth;

(3) audit the 3 and 4 product bundles, which show near-total attrition;

(4) run a dedicated retention plan for Germany, where churn is double that of France and Spain.

2. Background and Problem Statement

Customers can look financially strong, with a high balance or salary, and still leave because they are lightly engaged, hold few products or have a shallow relationship with the bank. Retention strategies

built on demographics or balance alone tend to be generic. The bank lacks quantitative evidence on (a) which behaviours drive retention, (b) whether product depth reduces churn, and (c) whether high balances alone secure loyalty.

3. Objectives

Primary: evaluate the engagement–churn relationship; measure the retention impact of product count and mix; identify disengaged yet high-value customers.

Secondary: support engagement-driven retention strategies; improve product bundling decisions; reduce silent churn among premium customers.

4. Data and Validation

The file has 10,000 rows and 14 columns: the 12 specified fields plus CustomerId and a Year column (all values 2025). Validation results:

Check	Result
Missing values	None in any column
Duplicate CustomerId	0
Binary fields (HasCrCard, IsActiveMember, Exited)	Only 0/1 values
NumOfProducts	1 to 4 (5,084 / 4,590 / 266 / 60 customers)
Geography	France 5,014; Germany 2,509; Spain 2,477
Churn label	7,963 retained, 2,037 exited (20.37%)
Active members	5,151 (51.51%)
Card holders	7,055 (70.55%)

Two data features matter for interpretation. First, 3,617 customers (36.2%) have a zero balance, and **every zero-balance customer is in France or Spain**: Germany has no zero balances. Second, the data is a single snapshot, so results show association, not causation. Balances are treated as euros for presentation, but the file does not state the currency.

5. Methodology

1. **Ingestion and validation** as above.
2. **Engagement classification** into four mutually exclusive profiles (Section 6). "High balance" means the top quartile, a balance of €127,644 or more.
3. **Product utilisation analysis:** churn by product count, single versus multi-product, and product count crossed with activity and geography.
4. **Financial commitment analysis:** balance bands, balance-by-activity, balance-to-salary ratio and mismatch groups.

5. **Retention strength assessment:** sticky-customer definition, stability across engagement tiers, and thresholds.
6. **KPIs** (Section 9) and a logistic regression as a robustness check that controls for geography, age and gender simultaneously.

6. Engagement and Churn

Group	Customers	Churn rate
Active members	5,151	14.27%
Inactive members	4,849	26.85%

6.1 Engagement profiles

Profiles are assigned in this order: inactive with high balance; other inactive; active with one product; active with two or more products.

Profile	Customers	Churned	Churn rate	Share of all churn
Active Engaged (active, 2+ products)	2,588	250	9.66%	12.3%
Active Low-Product (active, 1 product)	2,563	485	18.92%	23.8%
Inactive Disengaged (inactive, balance below top quartile)	3,602	922	25.60%	45.3%
Inactive High-Balance (inactive, top-quartile balance)	1,247	380	30.47%	18.7%

Inactive customers make up 48.5% of the base but account for **64% of all churn**. Even the active group is not safe if it holds a single product: Active Low-Product customers churn at about twice the rate of Active Engaged customers.

6.2 Activity effect holds in every segment

The gap between active and inactive customers appears in every country and for both genders:

Segment	Inactive churn	Active churn
France	21.13%	11.50%
Spain	23.35%	10.75%
Germany	41.08%	23.72%
Female	32.09%	18.13%
Male	22.28%	11.20%

In the logistic regression, which controls for age, geography, gender, balance, salary, tenure, credit score and card ownership, being an active member cuts the odds of churn by about two-thirds (odds ratio 0.33).

7. Product Utilisation

Products	Customers	Churned	Churn rate
1	5,084	1,409	27.71%
2	4,590	348	7.58%
3	266	220	82.71%
4	60	60	100.00%

- **Single versus multi-product:** single-product churn is 27.71%; multi-product (2 to 4) is 12.77%. This aggregate hides the real pattern, because the relationship is not linear.
- **Product depth is a curve, not a line.** Retention improves sharply from one to two products and then collapses. Customers with three or four products make up only 3.3% of the base but 13.7% of all churners (280 of 2,037).
- **The pattern is consistent across countries.** Two-product customers churn at 5.7% in France, 7.4% in Spain and 12.1% in Germany; customers with 3 or more products churn at 79% or more in all three.
- **Activity compounds product depth.**

	1 product	2 products	3 products	4 products
Inactive	36.65% (n=2,521)	9.89% (n=2,144)	88.24% (n=153)	100% (n=31)
Active	18.92% (n=2,563)	5.56% (n=2,446)	75.22% (n=113)	100% (n=29)

An inactive single-product customer is the single highest-risk large group (2,521 customers, 36.65% churn). Even inactive two-product customers (9.89%) are more loyal than active single-product ones (18.92%), which shows how strongly the second product anchors the relationship.

Interpretation of the 3 and 4 product result. The data cannot say why these customers leave. Plausible hypotheses to test with the bank's own records are product mis-selling or forced bundling, fee dissatisfaction, or service problems on complex bundles. This should be investigated before the bank pushes customers beyond two products.

8. Financial Commitment versus Engagement

8.1 Balance

Balance band	Customers	Churn rate
Zero	3,617	13.82%
€1 to 50,000	75	34.67%

Balance band	Customers	Churn rate
€50,000 to 100,000	1,509	19.88%
€100,000 to 150,000	3,830	25.77%
Above €150,000	969	23.12%

Customers with a balance churn at 24.08%, against 13.82% for zero-balance customers, so a higher balance is associated with *more* churn, not less. Part of this is a geography effect: Germany, the highest-churn country at 32.44%, has no zero-balance customers. After controlling for geography, age and the other factors, balance has almost no independent effect (odds ratio 0.94 per €100,000). The practical conclusion is that **balance is not a protective factor**: a large balance does not by itself make a customer loyal.

Churned customers held on average €91,109 versus €72,745 for retained customers, and churners held €185.6 million, or **24.3% of all balances**. Churn therefore carries a disproportionate financial cost.

8.2 Balance crossed with activity

	Inactive	Active
Balance below top quartile	25.60% (n=3,602)	13.42% (n=3,898)
Balance in top quartile (≥ €127,644)	**30.47%** (n=1,247)	16.92% (n=1,253)

High-balance customers are the most at risk precisely when they are inactive. Within the inactive high-balance group, single-product customers churn at 33.45% (n=828) and multi-product customers at 24.58% (n=419).

8.3 Salary and salary–balance mismatch

- Salary on its own barely moves churn: 20.00%, 19.80%, 20.12% and 21.56% across salary quartiles.
- The balance-to-salary ratio (for customers with a balance) is also flat, between 23% and 25% across all ratio bands. However, activity separates customers within every band: 30% to 33% churn for inactive and 15% to 18% for active.
- Mismatch groups: high balance with bottom-quartile salary (n=634) churns at 23.82%; low balance with top-quartile salary (n=1,238) churns at 17.12%. The first group is mildly riskier, but the difference is small. Salary mismatch is not a strong standalone risk signal in this data.

8.4 At-risk premium customers

Defined as inactive customers with a balance in the top quartile: **1,247 customers holding €185.6 million, with 30.47% churn.** The 867 who have not yet left hold **€128.8 million**. This is the silent-churn exposure the bank can still act on.

9. Retention Strength and KPIs

9.1 Sticky customer profile

Definition	Customers	Churn rate
Active and exactly 2 products	2,446	5.56%
...and age under 50	2,073	4.63%
...and age under 50 and not in Germany	1,583	3.85%

The sticky core is an active member with two products. Adding age and geography tightens it further, but the first two criteria carry most of the effect.

9.2 Key Performance Indicators

KPI	Definition	Result
Engagement Retention Ratio	Churn of inactive divided by churn of active	**1.88x** (26.85% / 14.27%)
Product Depth Index	Retention of each product tier relative to single-product retention (72.29%)	1 product: 1.00; 2 products: 1.28; 3 products: 0.24; 4 products: 0.00
High-Balance Disengagement Rate	Churn of inactive customers in the top balance quartile	**30.47%**
Credit Card Stickiness Score	Retention of card holders divided by retention of non-holders	**1.01** (79.82% / 79.19%), essentially no effect
Relationship Strength Index (RSI)	100 × product-tier retention × (0.5 + 0.5 × active flag)	See below

Relationship Strength Index. The RSI combines engagement and product depth into one 0 to 100 score. Churn by RSI band:

RSI band	Customers	Churn rate
0 to 20	326	85.89%
20 to 45	2,521	36.65%
45 to 70	2,144	9.89%
70 to 100	5,009	12.40%

The RSI ranks customers better than activity alone (AUC 0.72 versus 0.60). It is not perfectly monotonic: the top band mixes active two-product customers (5.6% churn) with active single-product ones (18.9%),

so it is best used to flag the bottom bands. The product-tier weights were calibrated on this same data, so the AUC is in-sample and slightly optimistic.

9.3 Engagement thresholds linked to retention

- **Activity:** being active roughly halves churn risk at every product level except the 3 and 4 product tiers.
- **Product count:** two products is the retention sweet spot; one is the main risk, and three or more is a red flag.
- **Age:** churn jumps from 12.1% (age 31 to 40) to 34.0% (age 41 to 50) and 56.2% (age 51 to 60), then falls to 24.8% above 60. The over-60 group is also the most active (80.8% active).
- **No threshold effect** appears for tenure (17% to 23% in every year), credit score (18% to 22% by quartile) or card ownership.

10. Robustness Check: Logistic Regression

A logistic regression with all fields (five-fold cross-validated AUC 0.83) confirms the direction of the main findings. Odds ratios for churn:

Factor	Odds ratio	Reading
Active member	0.33	Much lower churn odds
2 products (vs 1)	0.21	Much lower churn odds
3 or 4 products (vs 1)	16.6	Much higher churn odds
Germany (vs France)	2.58	Higher churn odds
Age, per 10 years	2.04	Higher churn odds
Male (vs female)	0.59	Lower churn odds
Credit card	0.94	No meaningful effect
Tenure, per year	0.98	No meaningful effect
Balance, per €100,000	0.94	No meaningful independent effect
Salary, per €100,000	1.05	No meaningful effect

Spain is similar to France (odds ratio 1.05). Behavioural variables (activity and products) remain strong after controlling for demographics, which supports the project's central premise. Age, geography and gender are also strong, so demographics are not irrelevant.

11. Recommendations

1. **Re-activate high-value inactive customers first.** Target the 867 retained inactive high-balance customers (€128.8 million) with relationship-manager calls, personalised reviews and engagement triggers (usage nudges, rate or fee reviews).

2. **Move single-product customers to a second product.** The biggest, most measurable lift is from one to two products (27.71% to 7.58% churn). Prioritise inactive single-product customers (36.65% churn). Target a second product, not more.
3. **Audit 3 and 4 product bundles before promoting them.** With 82.71% and 100% churn, these bundles may reflect bad fit, pricing or service. Interview or survey leavers and review sales practices.
4. **Deprioritise credit cards as a pure retention tool.** Card ownership shows no retention benefit; tie cards to engagement features such as rewards for regular use instead.
5. **Run a Germany-specific retention plan.** Churn is 32.44% (versus about 16% elsewhere), even among active customers (23.72%). Investigate pricing, competition and service locally.
6. **Tailor outreach by age and gender.** Customers aged 41 to 60 and women churn more, so life-stage offers and service reviews for these groups are justified.
7. **Adopt the RSI and sticky-customer flags as monitoring metrics,** refreshed each month, with alerts when a premium customer's activity lapses.
8. **Stop treating balance as a loyalty proxy.** Use engagement and product signals in segmentation instead.

12. Proposed Streamlit Dashboard Design

The analysis is structured so it maps directly to the required app modules:

- **Engagement vs churn overview:** churn by activity, profile and country; KPI cards for the Engagement Retention Ratio and overall churn.
- **Product utilisation impact:** churn by product count with activity and country splits; Product Depth Index.
- **High-value disengaged customer detector:** a filterable list of inactive customers above a chosen balance and salary threshold, with exposure total and churn risk.
- **Retention strength scoring:** RSI distribution, sticky-customer share and churn by RSI band.
- **Filters:** engagement status, product-count slider (1 to 4), balance and salary threshold sliders, plus geography and gender.

13. Limitations

- Single 2025 snapshot, so churn timing and cause cannot be observed; findings show association, not causation.
- IsActiveMember is a binary flag with no definition of how activity is measured, so finer engagement thresholds (logins, transactions) cannot be tested.
- Product type is unknown; only the product count is available, so "product mix" analysis is limited to count.
- Only 326 customers hold 3 or 4 products, so those rates are based on small groups, although the effect is very large.
- Zero balances exist only in France and Spain, which may reflect how the data was assembled; balance results should be read with this in mind.
- The currency is not stated and the RSI weights are calibrated in-sample.

14. Conclusion

Churn in this bank is shaped by relationship strength more than by wealth. Active membership and a second product are the strongest protective behaviours, balance offers no protection by itself, and the highest-value disengaged customers are both the riskiest large segment and the most actionable. Retention spend should therefore move from generic, balance-based campaigns towards targeted re-activation, careful bundling around two products, and an investigation of the three and four product bundles.

Appendix A. Executive Summary for Government Stakeholders

This analysis of 10,000 retail customers of a European bank examines why customers leave. About one in five (20.4%) left. Customers who do not actively use their accounts are almost twice as likely to leave as active ones, and those holding exactly two products are far more loyal than those with one. Large balances do not guarantee loyalty: inactive customers with the largest balances leave at the highest rate, and customers who left held a quarter of all deposits. The findings point to a need for better monitoring of customer relationships, for review of how multi-product bundles are sold, and for closer attention to customers in Germany. The analysis uses anonymised data and shows association, not proof of cause, so the recommendations should be piloted and measured before wide rollout.